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Executive Summary

Upon analyzing the nursing home industry, our team performed a variety of analytical tests and prediction models, including OLS Regression, Random Forest Analysis, Logistic Regression, as well as Lasso, Ridge, and Elastic Analysis, to better assess its financial capabilities and the quality of its services.

This report concludes that the nursing home industry is not suitable for financial investment because of its high-risk profile. Our team has analyzed provider finances and quality ratings to better understand how the industry operates, what are the most valuable factors to the industry, and where the industry could potentially be heading.

Introduction

Background Information: Objectives and Significance of the Analysis

The demand for elder care services has grown sharply in the United States due to the aging population, especially the baby boomer generation. As more people look for long-term care options, nursing homes are becoming increasingly attractive for private investment as well as public health planning. If the industry proves to be financially resilient and sustainable, particularly in the face of recent setbacks like the COVID-19 outbreak, this presents a potentially profitable opportunity for investors.

The purpose of this project is to determine if it would be beneficial for our customer to invest in nursing homes. In order to accomplish this, we will examine an extensive dataset covering nursing homes in the United States between 2015 and 2021. Thus, there will be five main objectives for the analysis:

1. Evaluate nursing facilities' overall financial performance nationwide,
2. Identify what important factors affected that performance,
3. Determine which of those elements affects events the most,
4. Describe long-term trends in performance and affecting factors,
5. Analyze the impact of the COVID-19 epidemic .

The significance of this analysis lies in its ability to provide insight into data-driven investment strategies in a sector that directly affects a vulnerable and expanding population segment.

Methodology

Data Collection

The data utilized in this research were obtained from the Centers for Medicare & Medicaid Services (CMS). They were made available via the Nursing Homes platform called the Provider Data Catalog and the Care Compare.

Data from several federal reporting systems is compiled by different sources, including (1) CMS Health Inspection Database, which contains inspection data, penalties, health deficiencies, and nursing home characteristics, (2) Payroll-Based Journal (PBJ) system, which provides comprehensive quarterly staffing records, (3) Minimum Data Set (MDS) National Database which performs clinical evaluations and quality metrics at the resident level, (4) Medicare Claims Data, which documents hospital stays, admissions, and outcomes related to quality.

The datasets utilized were in CSV format, arranged by year and data type, and covered the years 2015–2021. To ensure consistency across sources and years, we downloaded the data, combined and cleaned the datasets, about which we will talk in more detail in the next sections. The facility-level analysis of all datasets allowed for reliable cross-temporal and cross-geographic comparisons.

Data Analysis Techniques

Firstly, after looking at all of the yearly datasets, we merged them into unified datasets to make the working process easier. After that, we implemented the following methods of data analysis:

(1) Descriptive statistics: To find broad trends in financial performance and care quality over time, we started with simple statistical summaries. This involved figuring out important variables, including total discharges, operating expenses per discharge, and net income per discharge.
(2) Correlation analysis: We evaluated the connections between the financial results of different facility parameters (e.g., ownership type, staffing levels, quality ratings, and flaws). This helped draw attention to issues with multicollinearity.
(3) Trend & Time-Series Analysis: In order to detect major shifts in the industry, we studied the evolution of financial performance metrics over time, especially around the start of COVID-19, using visualizations and year-over-year comparisons.
(4) Regression Modeling (OLS): To identify the most statistically significant predictors of net income per discharge, we used Ordinary Least Squares regression with backward elimination. As a result, we were able to determine and analyze linear relationships.
(5) Categorical Modeling (Logistic, Decision Tree, Random Forest): We developed classification models that predicted whether a facility would receive a rating higher than three stars in order to evaluate the quality of the providers.

(6) Machine Learning: Random Forest & Regularized Regression (Lasso, Ridge, ElasticNet): To assess nonlinear correlations and feature importance in financial predictions, we implemented Random Forest analysis.
(7) COVID-19 Impact Assessment: In order to examine the effects of the pandemic on income, costs, and quality scores across areas, we separated COVID-era data (2020–2021). Systemic problems were also identified through region-specific analysis in places that were disproportionately impacted.

Analytical Tools / Software Used

To complete the project, we used a couple of industry-standard tools. At the very beginning, we used Excel for preliminary data review and manual cross-validation. The majority of the work was done using Python to take care of the data handling and visualizations.

Data Description

Overview of the Data Sets

As already mentioned, this project makes use of a wide range of datasets from the Centers for Medicare & Medicaid Services (CMS), particularly from the Provider Data Catalog and the Care Compare – Nursing Homes platform. Together, these datasets include information on the quality, operations, and financial aspects of nursing facilities in the US between 2015 and 2021.

The datasets are as follows:

(1) *ProviderInfo* includes facility details on nursing homes, such as facility name, address, ownership type, capacity, and certification status.
(2) *CostReport* gives data on financial performance and expenses, including total revenue, operating expenses, profit margins, and capital costs.
(3) *HealthDeficiencies* provides information on inspection violations and deficiency tags, categorized by type, severity, and date.
(4) *QualityMsrMDS* gives metrics for the quality of care obtained from MDS evaluations, including but not limited to mobility decline and infection rates.
(5) *Penalties* include information about fines and other enforcement measures that noncompliant facilities face.
(6) CovidVax provides staff and resident vaccination rates for 2020 and 2021.

In addition to being arranged by year, each dataset was supplied in CSV format and included a *DataDictionary* that offers thorough explanations of every variable.

The combined dataset represents thousands of distinct nursing home facilities with hundreds of thousands of entries spanning all years and tables. A variety of data types are included in the

variables, including date/time, numeric (such as total costs, staffing hours), and categorical (such as ownership type, state).

Overall, finding trends, patterns, and important performance drivers in the nursing care sector is made much easier by the wide range and variety of this data.

Data Preprocessing

The raw datasets were carefully preprocessed before any analysis started in order to guarantee accuracy, consistency, and usability across several sources and years.

Cost Reports

The Cost report files were arranged by year and category, and each dataset from 2015 to 2021 was given its own CSV file. Each set of annual files was combined into a single dataset using Python and the pandas package. In order to preserve each record's temporal context, a new column called "Year" was added during this phase. In many cases, different years or datasets had different column names, requiring harmonization to match naming conventions and allow smooth merging. Once this was done, we added new columns that aggregated much of the financial data. These new columns were the sums of total_discharges_total, Total_Discharges_Title_XVIII, Total_Discharges_Title_XIX, and Total_Discharges_Title_Other. The most important of these new columns is **Net_Income_per_Discharge**. We used discharges as a proxy for patient counts because there was no count of patients in the original datasets. Discharges are patients who have completed whatever treatment/allotted time. This new column was used as the target variable for many of the future models.

To improve model accuracy and reduce bias, the preprocessing started by eliminating all rows that lacked the target variable (Net_income_per_discharge). Any row with more than 50% of the independent variables' values missing was eliminated to control the quality of the data. This made sure that only records with enough information were used to train the model. For the remaining data, to impute numerical missing values, we used the median of each corresponding column, which is a reliable technique that is less prone to outliers. And to impute categorical missing values, we used the mode, guaranteeing the dataset's logical consistency. However, we understood that categorical variables do not add much value to the analysis in this case, and that’s why we later deleted the columns containing categorical variables. Furthermore, to avoid model distortion and make sure every data point reflected a distinct observation, duplicate records were eliminated.

To prevent type-related problems during modeling, we transformed all variables to numeric types after the features and target were established. Then, we used Scikit-learn's StandardScaler to standardize the resulting feature matrix, converting the variables to have a unit variance and zero mean. Standardization was crucial for the Ordinary Least Squares (OLS) regression to avoid variables with larger values disproportionately affecting the model.

This preprocessed dataset was fed into two different modeling techniques: a Random Forest regression model for non-linear insights and variable importance analysis, and backward elimination using linear regression.

We used a composite key that consisted of rpt_rec_num, provider_ccn, and Year. This connects the datasets Provider Information and Cost Reports. We were able to conduct a thorough examination of financial performance alongside operational and quality measures.

Provider Info, Health Deficiencies, Quality Measurements, Penalties

All remaining files provided concerning provider information (including the CSV files for Provider Info, Health Deficiencies, Quality Measurements, Penalties) contain important categorical data about the nature of the nursing home industry. The process to clean and prepare these files was centered around making them usable for categorization models, assessing the quality of care provided.

Each CSV, much like the cost reports, was divided into the previously mentioned categories and then by year. Our team, prior to performing any processing, assessed the dictionaries for the datasets and determined which columns within each were of interest and worth keeping.

Provider Information

The first step in processing the data collection was inserting a column in each CSV with the year of the file (ex, ProviderInfo_2015.CSV has a new column, [Year] with data value = “2015”). It was also noted that the COVID-19 years, 2020-2021, had more columns and different names for parallel columns. The next step was to adjust the column titles so that the formatting was consistent for both coding and data preparation (in this case, selecting columns with desired data) purposes; the new columns of the COVID-19 years CSV files would be addressed later. It was decided that in order to properly measure the potential differences between the pre-COVID-19 years CSV files and the COVID-19 years CSV files, any collation needed to first produce two files for separate analysis (one for the range 2015-2019 and the other for the range 2020-2021) for each category. With all the categories collated around the year range, the next step was to select and begin modifying the base file, the anchor to which the other files would join when merging.

The CSV files for Provider Information were the perfect candidate(s) for the base file, setting the primary keys for all the data sets as provider federal number [Provnum] and year of the instance [Year]. The first task was to select the previously determined relevant columns to keep. The next step was an assessment to find any null values and determine the distributions of the columns, if it was necessary to fill the nulls. Upon viewing the nulls reports,[1] and in proportion to the total

[1] [Ap.1][Ap.2]

size of the data sets, a maximum of 3016 nulls (3.87%) within a total of 78025 data values in the PreCOVID-19 files and a maximum of 4545 nulls (14.85%) within a total of 30612 data values in the COVID-19 files being removed would not disrupt the distributions of the existing columns. IQR was also applied to all numeric values to reduce the number of outliers in the data, which brought down the data value counts to 32779 and 11033, respectively. This reduction of data values in the interim is recognized as being meaningfully substantive and potentially a problem for properly assessing the ability of the data set to properly reflect information about the providers. This deduction is important because while the distributions were more or less normalized, any risk of outliers reducing the effectiveness of the model must be reduced. Later on, before the completion of the data preparation, more instances will be added as means to cover for missing information.

Deficiencies

The next datasets to prepare were the Deficiencies.[2] which marks the type of citations a provider will receive for issues that the provider will have (i.e., fire safety, health procedures, health inspection failures, etc.). This dataset does have a collection of data that might be important if assessing the capabilities and efficacy of a particular provider, however, for our team's purposes, there is a need for reduction. Functionally, an investor at this point in time trying to better understand the industry only needs to understand how many times this is an occurrence and if the quantity of these citations has been increasing over time. That is why it was decided to create a new column [DeficiencyCount], which aggregated the number of times a deficiency occurred for a provider within a year. This was performed via a loop that used [deficiency tag] to form the counts. What remained was the composite primary key and [DeficiencyCount]. This was then inner-joined with the previous provider info dataset, producing a new base file. When merging, the data types for the composite keys need to match, and so they are casted as strings while the merge occurs.

Quality Measurement

Centers for Medicare & Medicaid Services (CMS) performs an assessment of each provider every year and calculates their measure scores and determines the measure average.[3] Measures are incidents of note, both positive and negative, that occur and are reported during a year. Amongst the variables within the dataset, the most important are the measure descriptions/code, the stay type, the measure scores, and the measure average.

There are multiple instances a provider can have per year, and CMS measures each of them. Given the way we wanted to structure our dataset with the composite keys and how we wanted to conduct analysis, the number of rows had to be reduced so that there was only one instance per provider per year. The way to do this was by means of consolidating, in particular, combining both [stay_type] and [measure_description]. Running through a for loop, it would assess the stay type of the instance and delegate each stay into its own category. Then, the

[2] CMS-ProviderData
[3] CMS-QualtyMeasures

measure scores for a provider for a year would be summed, and the average would be calculated. If this average was greater than a preset threshold of 20, it would be considered a positive measure or else not (marked by adding to a count). With these aggregations performed, the file would be inner-joined again to the base file.

Penalties

Providers incur penalties and fines depending on the nature of the deficiencies. Similar to the deficiencies, all that is needed for an investor to understand is the number of penalties a provider incurs and the dollar amount of fines a provider has incurred within a year. A for loop performs both the counting of the penalties and the summing of the fines. Lastly, the file is merged with the base file. In this instance, though, the pre-COVID-19 files did not provide enough material to use in predictive modelling, so the information was removed.

Final Data Preparation

With all the category files combined into one, each for both ranges, the final data preparation can be made to help better assess the provider information. The team decided it was best to create dummy variables for the following columns:

- [ownership_type] = data was reduced to only include government, Nonprofit, and for-profit
- [CERTIFICATION] = includes Medicaid, Medicare, and both
- [ZIPREGION] = reduction of Zip Code to its region (first digit)
- [Score_High] = **Target Variable**

Centers for Medicare & Medicaid Services (CMS) include a rating system that scores providers each year on a scale from 1-5. The target variable needs to be a binary which assesses whether or not a provider scores high; in this case, a score above 3 is high (1 = True, 0 = False).

The final step was that there were some instances that were missing, specifically years for each provider (in particular, the pre-COVID-19 years), which needed imputation. Using a for loop grouping by provider federal number, the provider years would be checked against the range included in a list and if the year did not exist it would be filled with the median values of that provider (ex: Pr#15014 is missing two years from the range, inserts two rows for the missing years with values that correspond with the median values for that provider).

Thus, the dataset is ready for categorical modeling and analysis.

Diagnostics

Prior to moving on to more complex modeling, we first assessed the overall financial performance of U.S. nursing facilities between 2015 and 2021. Standardized net income per

discharge (Z-score) and average operating expense per discharge (in USD) are two important metrics that we compared in Appendix 15. According to the analysis, net income showed a rather steady rising trend through 2018, but starting in 2019, when COVID-19-related factors began to increase, it started to decrease sharply. Operating costs also showed a declining trend over the same period, most likely as a result of both operational reductions and a decrease in patient volume. The steady correlation between these two patterns implies that, particularly during and after the pandemic, financial performance has become more unpredictable and less sustainable in recent years.

Analysis and Findings

OLS Regression Analysis

We performed a multiple linear regression analysis with Net Income per Discharge as the dependent variable to identify the primary financial factors that influence nursing home performance. By normalizing net income across facilities with different sizes and discharge volumes, this statistic offers a fair comparison point. Initially, a total of 24 independent variables were chosen, covering a broad variety of operational and financial measures such as total revenue, different discharge types, patient income streams, cost breakdowns, and financial indicators for each discharge.

To identify the most statistically significant factors, we used a backward elimination process, which would also refine the model. In this iterative procedure, p-values were used to eliminate the least significant variables one by one from each of the 24 features until all of the remaining predictors had p-values below the significance level of 0.05. Three predictors that didn't reach the significance level were eliminated using backward elimination, giving us a somewhat neater model with 21 independent variables. The remaining factors have a statistically significant relationship with Net Income per Discharge, as indicated by p-values < 0.05.

To dive deeper into the results, with a final linear regression model's R-squared value of 0.545, the independent variables in the model account for roughly 54.5% of the variance in net income per discharge. In other words, this indicates that the variables that we have modeled explain slightly more than half of the changes in nursing facilities' financial performance. Despite its shortcomings, this suggests an acceptable level of explanatory power, which is especially significant considering the complex financial aspects of nursing home operations. In other words, each feature that was kept plays a major role in forecasting financial performance. The overall significance of the model is confirmed by the p-value near 0 and the F-statistic of 5927, which determines if our model fits the data more accurately than a model that only uses the target variable's mean as a predictor. See Appendix 16.

Overall, the OLS regression model found several influential variables that have a big impact on nursing homes' financial performance. Revenue per discharge was the most influential positive

variable, suggesting that institutions with more patient revenue are more profitable. On the other hand, there was a significant negative impact on operating expense per discharge, emphasizing the importance of cost control. Other significant factors that emphasized the significance of achieving a balance between income and expenses were gross revenue (positive), total costs, and staff wages (both negative). Furthermore, facilities with larger discharge volumes also tended to do better financially, indicating that operational effectiveness and size play a significant role in profitability.

Categorization Model (Logistic Regression, Decision Tree, Random Forest) Analysis

As discussed before, our target variable to predict is scoring above a 3 on the provider's overall rating. For our purposes, the most important element outside of understanding how the industry is financially performing is to understand the quality of this industry, which can be determined by referencing the overall ratings. **An investor would be interested in understanding which factors are the most significant when contributing to the rating of the provider**.

Correlation Coefficient Analysis

For the first assessment, the correlation coefficients are assessed to both understand which variables might best correspond with the target as well as mark which variables are multicollinear. Upon viewing both correlation matrices,[4] while the scores signifying meaningful correlation relationships (defined by scores between .25 and .75) between [Score_High] and the rest of the variables were not incredibly high, this does not automatically mean there are no variables best built to help predict higher overall ratings.

The variables for the PreCOVID-19 that have the highest correlation coefficient scores and relevancy to predicting the target are as follows:[5]

- ["quality_rating"] = .68
- ["staffing_rating"] = .39
- ["DeficiencyCount"] = -.43
- ["adj_total"][6] = .28

These correlation coefficients will also help in the modelling process when determining which variables are selected.

Categorization Models

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[4] [Ap.3][Ap.4]
[5] [Ap.5][Ap.6]
[6] To note the collection columns showing hours worked by staff per resident has the most multicollinear variable relationships between themselves.

Prior to performing the Logistic Regression, Decision Tree, and Random Forest models, the variables needed to be selected via a selection process. Upon initially performing the categorization models with the forward selection method, the model's Accuracy/Precision/Recall/F scores were too high to be considered reliable (range: .98-1.00). A manual selection process relying on both assessment of the correlation coefficient scores as well as industry insight was used; in particular the two rating metrics (quality and staffing) which denote satisfaction with the service, the deficiency count for how many times negative issues appeared, and the collective total number of hours staff spends with a resident serving as a quantifiable measurement of labor needed to have a high overall rating.

However, when assessing the COVID-19 variables and their correlation coefficient scores, it was clear that what correlated more (negatively) with Score_High were the Zip Code regions. This was taken into consideration when performing the categorization models.

Many model tests were performed separately on the two datasets, each with varying results and requiring adjustments to each dataset. For instance, while the scores for predicting class 1 were decent (accuracy: LR = .79, DT = .85, RF = .87, Precision: .80-.88, Recall: .82-.88, F-score: .81-.88), the PreCOVID-19 required more columns to produce a better sensitivity curve when graphing the ROC. The following columns were added: ["short_stay_poor_quality"], ["short_stay_good_quality"], ["long_stay_poor_quality"], ["long_stay_good_quality"], ["measure_score_qtravg"]. This brought the model scores up (accuracy: LR = .80, DT = .84, RF = .88, Precision: .80-.89, Recall: .84-.91, F-score: .82-.90)[7] and fixed the curve(s).[8]

Next, the COVID years dataset was put through the models, first using the variables with the highest correlation coefficient scores. This resulted in all scores being equal to 1.00. It was then decided to use the first model's variables to make a more accurate prediction model, which netted more usable results (accuracy: LR = .71, DT = .79, RF = .85, Precision: .74-.88, Recall: .79-.87, F-score: .76-.87).[9] The model scores, the ROC curve, and the Precision/Recall curve visuals showed that the model was not as accurate, but still was able to function well enough to confirm the independent variables' ability to predict the target.[10]

As a result, it is clear that the main variables (quality rating, staffing rating, deficiency count, total hours per resident) were important in determining how well a provider performs its services. From here, the datasets can be merged with the cost report information for further analysis.

Another functional purpose of these categorization models was to assess whether or not the variables within the model would operate as one would expect before merging this with the cost report file. The best way to ensure this was to confirm the ability of the variables to predict the quality of each provider.

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[7] [Ap.7]
[8] [Ap.8]
[9] [Ap.9]
[10] [Ap.10]

Random Forest Analysis

As a supplementary modeling technique to the linear regression (OLS), we carried out the Random Forest analysis to determine the most important factors influencing nursing home financial performance, particularly Net Income per Discharge. While OLS regression presumes a linear relationship between predictors and the target variable, Random Forest is an ensemble-based, non-parametric machine learning technique that may identify complex interactions, nonlinear patterns, and the significance of variables in high-dimensional data. To increase forecast accuracy and manage overfitting, Random Forest constructs numerous decision trees on random subsets of the data and averages their outputs. This makes it particularly useful in real-world financial statistics like these, where there may not be a strict linear trend in the correlations between inputs (such as costs, discharges, and staffing) and outcomes (like net income).

The preprocessed dataset used for linear regression was also used in this analysis. We created a training set (80%) and a testing set (20%) from the data after the feature matrix was standardised. Next, a fixed random state for repeatability and 100 estimators (trees) were used to train a RandomForestRegressor from scikit-learn.

The Random Forest regression model obtained an R-squared score of 0.9508, meaning that it can account for almost 95% of the variation in net income per discharge. This model produced a very high R-squared, which seemed impressive at first glance. To compare real and expected net income per discharge, we also generated a scatter plot. There were a few extreme outliers in the data, which indicated that some facilities displayed financial patterns that the model was unable to adequately account for, which might include unmeasured factors such as management quality. To prevent the model distortion, we masked the outliers, and as a result, it became clear from the dense clustering along an upward line that the Random Forest model is doing a good job of capturing the overall direction and structure of the financial performance data. But looking at the significance scores and the almost perfect diagonal scatter plot, we understood that it could be because the dependent variable was defined or overlapped with some of the factors, including Total Income and Net Income from Patients, which were direct components of the target variable, Net Income Per Discharge. As a result, the model was less effective at identifying independent performance factors since Total_Income, which was the leading predictor, had a dominant significance score of 0.67. An additional measure of the average predicted error is the mean squared error (MSE), which represents the average of the squared differences between the actual and predicted values. In our case, MSE is around 2.07 million. This is more interpretable in a real-world financial context when converted to Root Mean Squared Error (RMSE), which indicates an average prediction error of about $1,439 each discharge. See Appendix 17 for the Importance Analysis and the Scatterplot.

In order to address the problem of including direct components in the analysis, we ran another version of the model after eliminating these variables, which produced a more realistic R-squared of 0.5299. Despite a decline in forecasting power, the model became more reliable for detecting important features. As a result, none of the features was overly dominant.

Net_Income_from_patients, Revenue_per_discharge, and SNF_Discharges_Total became the top features. The new scatter plot showed a more natural spread, which still shows a positive correlation between actual and predicted values. We believe that this method produced a model that offers more reliable insights for strategic decision-making by avoiding inflating model performance by eliminating elements that naturally characterize the result. See Appendix 18 for the Importance Analysis and the Scatterplot.

Lasso, Ridge, and Elastic Analysis

The Lasso Regression was performed on the combined cost report (all years merged) and provider information CSVs.

<u>Key Takeaways from the Lasso Regression:</u>
The variables were split between those that had a positive impact on net income per discharge and those that had a negative impact.

Positive variables include CERTIFICATION_Medicare(a), RESTOT (+830)(b), Score_High (+461)(c), ZIPREGION_2 (+208) & ZIPREGION_9 (+234)(d), short_stay_good_quality (+397)(e), staffing_rating (+302)(f), and overall_rating (+273)(g). Variable A shows that medicare certification leads to higher income. B is simply the total number of residents, but it does show that more residents mean more income by a good margin. C shows that higher facility scores mean more income per discharge. It is interesting that it is higher than the regions. Speaking of regions, Zip regions 2 and 9 show that region does matter, but that it is probably due to those locations' relative economic wealth. Variable E shows that high-quality short-term care is associated with better financial performance. Longer stays may mean expenses begin to outweigh the revenue accrued. The staff rating, variable F, has a greater impact than location. Better staff means higher customer satisfaction and likely repeat customers, and more new ones. The overall rating, variable E, shows that better-rated facilities generally make more per discharge.

Negative variables include BEDCERT (−971)(a), adj_total, adj_rn, adj_lpn (−600s to −270s)(b), DeficiencyCount (−485)(c), incident_cnt & cmplnt_cnt (−60 to −170)(d), easure_score_qtravg & WEIGHTED_ALL_CYCLES_SCORE (−123 to −147)(e), OWNERSHIP_Government (−113)(f), and Year (−113)(g). Variable A, Bedcert, shows that the more beds the facility has, certified it means less income. Perhaps more beds mean a higher operating cost. There is also the issue of more beds needing more space, which will increase rent/ownership costs. Variable B shows that adjusted staffing hours seem to represent higher labor costs without increasing income. The cost/income ratio per staff member seems to become unbalanced the more staff there are. DeficiencyCount, variable C, shows that more deficiencies mean lower income. This probably reflects penalties or poor and inefficient management. Incident counts and complaints (d) are very similar to the previous variable, but specific to incidents and complaints. Variable E shows that, again, bad scores mean lower profitability. Potential customers are likely looking elsewhere.

Whether or not the location is run privately or by the government is relevant, too. Government-run homes tend to earn less per discharge. Finally, the year (g) variable shows that net income per discharge decreased over time. COVID-19 certainly has an impact here.

Discussion

Interpretation of the Findings

Trends and COVID-19

Number of stays (good/bad) per year: The absolute number of stays remains steady until 2019 and especially 2020, when it drops significantly. It rises back up very slightly in 2021 but is still well below pre-COVID norms. Also, the number of good stays far outweighs the number of bad ones. (Appendix 11)

Net Income per Discharge (standardized): We took total net income and divided it by total discharges to get the time series trend. It shows that the income increases moderately till 2019 in when it begins to drop dramatically. It is interesting to see that the income drop trend continues at the same rate from 2019 to 2020. Perhaps this is because COVID-19 technically began in late 2019; we often forget this and associate its beginning as being in 2020.
(Appendix 12)

Net Income per Discharge per ZIP Region (standardized): With one outlier, the regions are all similar until COVID happens. It causes incomes to drop dramatically and then rise back up. It may appear that the incomes approach normality, but due to the standardization of the income numbers, we cannot assume that.
(Appendix 13)

Operating Expenses per Discharge (standardized): The expenses trend down slightly before Covid and then plummet during. This could be because of fatalities and a lack of new patients.
(Appendix 14)

In regard to the categorical analysis of quality of care provided by nursing homes, both the pre-COVID-19 dataset and the COVID-19 dataset (even though the correlation coefficients for the end independent variables were lower) were able to (generally) make similar predictions. This means that the ratings are still properly affected by the previously mentioned influential factors.

One thing to be observed was that before COVID-19, the region where providers reside did not meaningfully affect the ratings. Once COVID-19 occurred, regions played a bigger role in determining how good the quality of care would be. Given that the correlation coefficients were negative, this could indicate which region of nursing homes was not better prepared to deal with

COVID-19 than others. In turn, this also potentially reveals which are the broader weak point regions of nursing homes, regions that don't operate as well as others (either through lack of resources or clients).

Limitations and Considerations

With a limited amount of time available, there were only so many models to run and test and provide meaningful insight into. A complete breakdown of each region and then each state using regression and categorical analysis would be good for evaluating performance on a fractal level. This first would require the investor to still be interested in investing in the nursing home industry, and secondly, enough time to not just run the models alone but also perform more research into each region's provider ecosystems.

From an analysis perspective, there were limitations concerning the data itself. While there were numerous amounts of data, much of it had to be engineered to fit the needs of the model to perform better (ex, for missing year instances per provider, the median values of that provider that year were used for imputation). There is also another categorization model that needs to be tested, determining which variables could help predict whether a provider can achieve a profit margin (for potential growth).

It would also be interesting to perform a likewise industry analysis of an adjacent field to compare results and potentially find an alternative to investing in nursing homes.

Recommendations

We can conclude, given our data, that it would be unwise to invest in this sector. The net income per discharge doesn't take into account how many discharges there are in total, so it becomes evident that the profitability of this sector has declined dramatically far past its previous inability to make a profit. It also varies by region, mostly down. It is rather volatile and the margins look too small.

Conclusion

In short, it was determined that the industry was not suitable for investment as it was not financially stable due to its general inability to have a strong profit margin. Its precarious business model, which, based on income from its residents, would not be able to sustain itself due to the multiplying expenses incurred per additional resident. The quality of each nursing home region has only worsened over time. These reasons alone make this a poor investment option, not considering the disastrous effects of COVID-19, which compounded already existing issues.

Appendices

1) PreCOVID-19 | Provider Info Null Values

```
Null Values in Each Column:
ï»¿provnum                    0
Year                          0
PROVNAME                      0
ADDRESS                       0
CITY                          0
STATE                         0
ZIP                           0
OWNERSHIP                     1
BEDCERT                       0
RESTOT                      588
CERTIFICATION                 0
INHOSP                        0
overall_rating             1252
survey_rating               808
quality_rating              949
staffing_rating            2338
RN_staffing_rating         2338
AIDHRD                     2699
VOCHRD                     2699
RNHRD                      2699
TOTLICHRD                  2699
TOTHRD                     2699
PTHRD                      2708
exp_aide                   1702
exp_lpn                    1702
exp_rn                     1702
exp_total                  1702
adj_aide                   3016
adj_lpn                    3016
adj_rn                     3016
adj_total                  3016
WEIGHTED_ALL_CYCLES_SCORE     0
incident_cnt                  0
cmplnt_cnt                    0
FINE_CNT                      0
```

```
Percentage of Nulls:
ï»¿provnum                    0.00
Year                          0.00
PROVNAME                      0.00
ADDRESS                       0.00
CITY                          0.00
STATE                         0.00
ZIP                           0.00
OWNERSHIP                     0.00
BEDCERT                       0.00
RESTOT                        0.75
CERTIFICATION                 0.00
INHOSP                        0.00
overall_rating                1.60
survey_rating                 1.04
quality_rating                1.22
staffing_rating               3.00
RN_staffing_rating            3.00
AIDHRD                        3.46
VOCHRD                        3.46
RNHRD                         3.46
TOTLICHRD                     3.46
TOTHRD                        3.46
PTHRD                         3.47
exp_aide                      2.18
exp_lpn                       2.18
exp_rn                        2.18
exp_total                     2.18
adj_aide                      3.87
adj_lpn                       3.87
adj_rn                        3.87
adj_total                     3.87
WEIGHTED_ALL_CYCLES_SCORE     0.00
incident_cnt                  0.00
cmplnt_cnt                    0.00
FINE_CNT                      0.00
```

2) COVID-19 | Provider Info Null Values

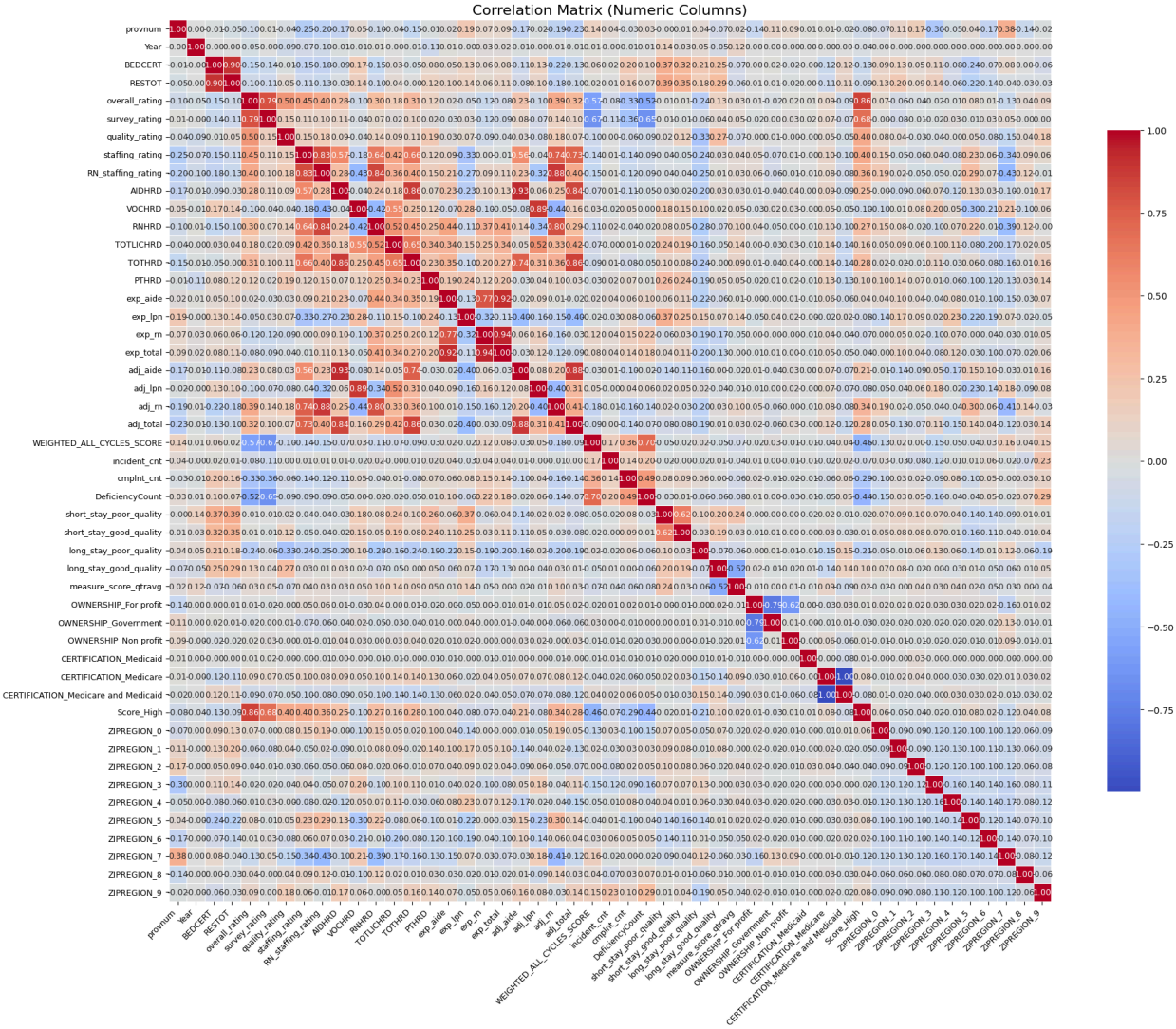
```
Percentage of Nulls:
Federal Provider Number                                    0.00
Year                                                       0.00
Provider Name                                              0.00
Provider City                                              0.00
Provider State                                             0.00
Ownership Type                                             0.00
Number of Certified Beds                                   0.00
Average Number of Residents per Day                        0.44
Provider Type                                              0.00
Overall Rating                                             1.44
Health Inspection Rating                                   1.44
QM Rating                                                  1.63
Long-Stay QM Rating                                        4.17
Short-Stay QM Rating                                      14.85
Staffing Rating                                            5.73
RN Staffing Rating                                         5.73
Adjusted Total Nurse Staffing Hours per Resident per Day   4.41
Total Weighted Health Survey Score                         0.87
Number of Facility Reported Incidents                      0.00
Number of Substantiated Complaints                         0.00
dtype: float64
```

```
Null Values in Each Column:
Federal Provider Number                                    0
Year                                                       0
Provider Name                                              0
Provider City                                              0
Provider State                                             0
Ownership Type                                             0
Number of Certified Beds                                   0
Average Number of Residents per Day                      135
Provider Type                                              0
Overall Rating                                           440
Health Inspection Rating                                 440
QM Rating                                                500
Long-Stay QM Rating                                     1277
Short-Stay QM Rating                                    4545
Staffing Rating                                         1755
RN Staffing Rating                                      1755
Adjusted Total Nurse Staffing Hours per Resident per Day 1350
Total Weighted Health Survey Score                       267
Number of Facility Reported Incidents                      0
Number of Substantiated Complaints                         0
dtype: int64
```

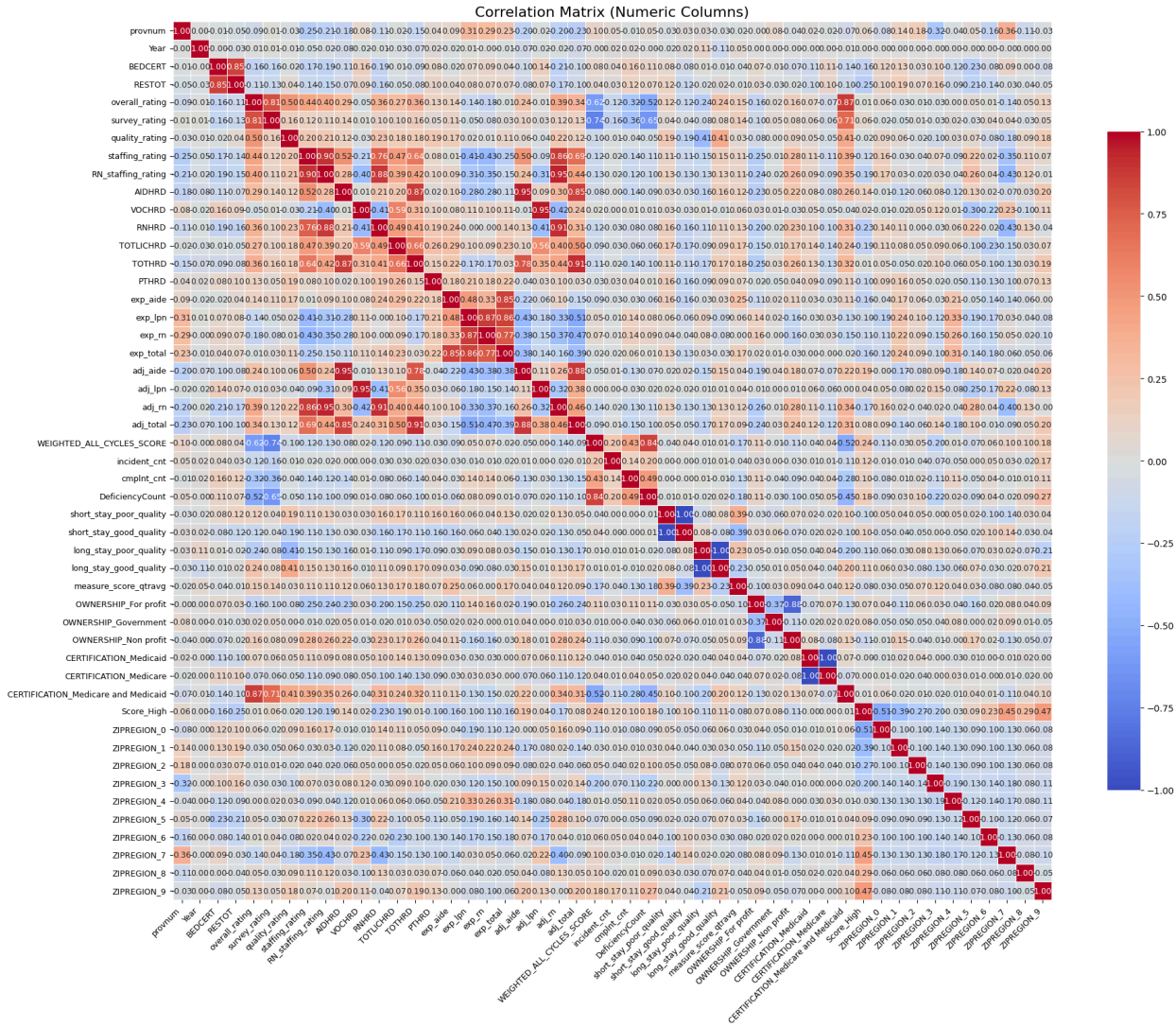
3) <u>Correlation Coefficients | PreCOVID-19</u>

```
Top 20 variables most correlated with Score_High:

overall_rating               0.858556
survey_rating                0.680999
WEIGHTED_ALL_CYCLES_SCORE   -0.461170
DeficiencyCount             -0.439284
staffing_rating              0.397996
quality_rating               0.397789
RN_staffing_rating           0.359251
adj_rn                       0.339699
cmplnt_cnt                  -0.286377
adj_total                    0.282457
TOTHRD                       0.275364
RNHRD                        0.273706
AIDHRD                       0.251534
adj_aide                     0.213447
long_stay_poor_quality      -0.209689
TOTLICHRD                    0.159293
BEDCERT                     -0.134627
ZIPREGION_7                 -0.117141
long_stay_good_quality       0.103834
VOCHRD                      -0.099835
Name: Score_High, dtype: float64
```

4) Correlation Matrix | PreCOVID-19



5) <u>Correlation Matrix | COVID-19</u>



6) <u>Correlation Coefficients | COVID-19</u>

```
Top 20 variables most correlated with Score_High:

ZIPREGION_0                 -0.509066
ZIPREGION_9                  0.474365
ZIPREGION_7                  0.452450
ZIPREGION_1                 -0.392525
ZIPREGION_8                  0.287705
ZIPREGION_2                 -0.266814
RESTOT                      -0.247882
WEIGHTED_ALL_CYCLES_SCORE    0.239483
ZIPREGION_6                  0.232752
RNHRD                       -0.231082
ZIPREGION_3                 -0.203025
adj_aide                     0.193016
RN_staffing_rating          -0.187922
TOTLICHRD                   -0.186849
DeficiencyCount              0.180125
adj_rn                      -0.166520
exp_aide                    -0.161106
exp_total                   -0.159161
BEDCERT                     -0.157141
AIDHRD                       0.140454
Name: Score_High, dtype: float64
```

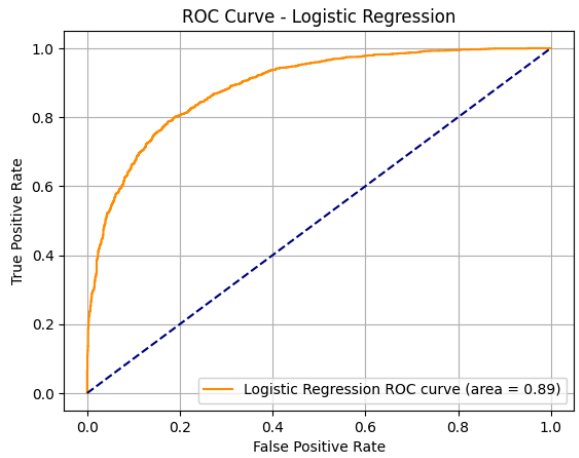
7) <u>2018-2019 Categorization Model Results</u>

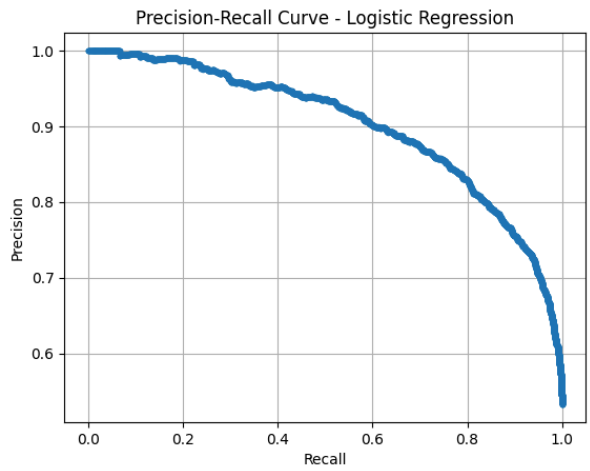
```
=== Logistic Regression ===
Accuracy: 0.8018999073215941
Confusion Matrix:
[[1538  478]
 [ 377 1923]]
Classification Report:
              precision    recall  f1-score   support

           0       0.80      0.76      0.78      2016
           1       0.80      0.84      0.82      2300

    accuracy                           0.80      4316
   macro avg       0.80      0.80      0.80      4316
weighted avg       0.80      0.80      0.80      4316


=== Decision Tree ===
Accuracy: 0.8403614457831325
Confusion Matrix:
[[1659  357]
 [ 332 1968]]
Classification Report:
              precision    recall  f1-score   support

           0       0.83      0.82      0.83      2016
           1       0.85      0.86      0.85      2300

    accuracy                           0.84      4316
   macro avg       0.84      0.84      0.84      4316
weighted avg       0.84      0.84      0.84      4316


=== Random Forest ===
Accuracy: 0.8880908248378128
Confusion Matrix:
[[1750  266]
 [ 217 2083]]
Classification Report:
              precision    recall  f1-score   support

           0       0.89      0.87      0.88      2016
           1       0.89      0.91      0.90      2300

    accuracy                           0.89      4316
   macro avg       0.89      0.89      0.89      4316
weighted avg       0.89      0.89      0.89      4316
```

8) <u>2020-2021 Categorization Model Results</u>

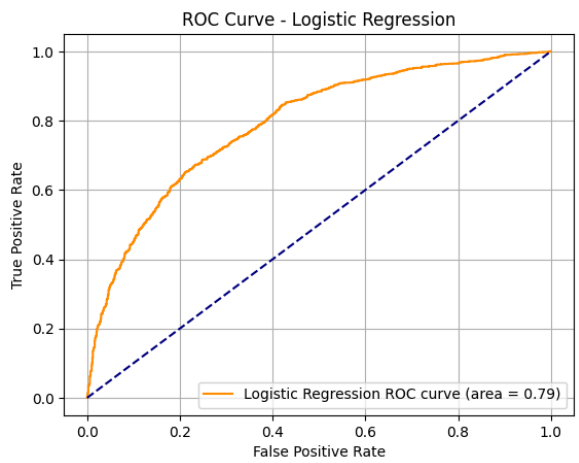
```
=== Logistic Regression ===
Accuracy: 0.7198483804272915
Confusion Matrix:
[[ 777  460]
[ 353 1312]]
Classification Report:
              precision    recall  f1-score   support

           0       0.69      0.63      0.66      1237
           1       0.74      0.79      0.76      1665

    accuracy                           0.72      2902
   macro avg       0.71      0.71      0.71      2902
weighted avg       0.72      0.72      0.72      2902


=== Decision Tree ===
Accuracy: 0.7925568573397657
Confusion Matrix:
[[ 938  299]
[ 303 1362]]
Classification Report:
              precision    recall  f1-score   support

           0       0.76      0.76      0.76      1237
           1       0.82      0.82      0.82      1665

    accuracy                           0.79      2902
   macro avg       0.79      0.79      0.79      2902
weighted avg       0.79      0.79      0.79      2902


=== Random Forest ===
Accuracy: 0.8559614059269469
Confusion Matrix:
[[1032  205]
[ 213 1452]]
Classification Report:
              precision    recall  f1-score   support

           0       0.83      0.83      0.83      1237
           1       0.88      0.87      0.87      1665

    accuracy                           0.86      2902
   macro avg       0.85      0.85      0.85      2902
weighted avg       0.86      0.86      0.86      2902
```

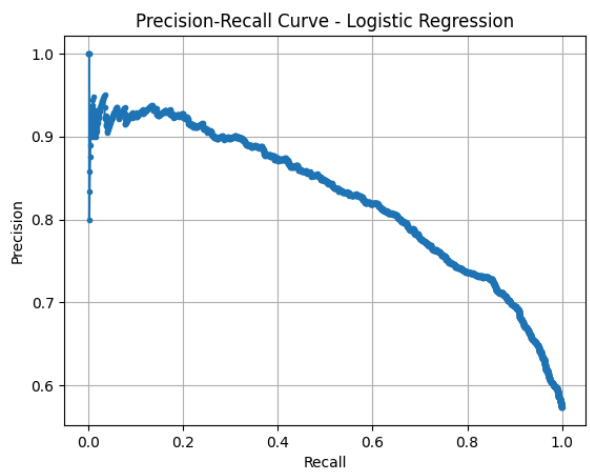
9) <u>PreCOVID-19 Categorization Model Visuals</u>



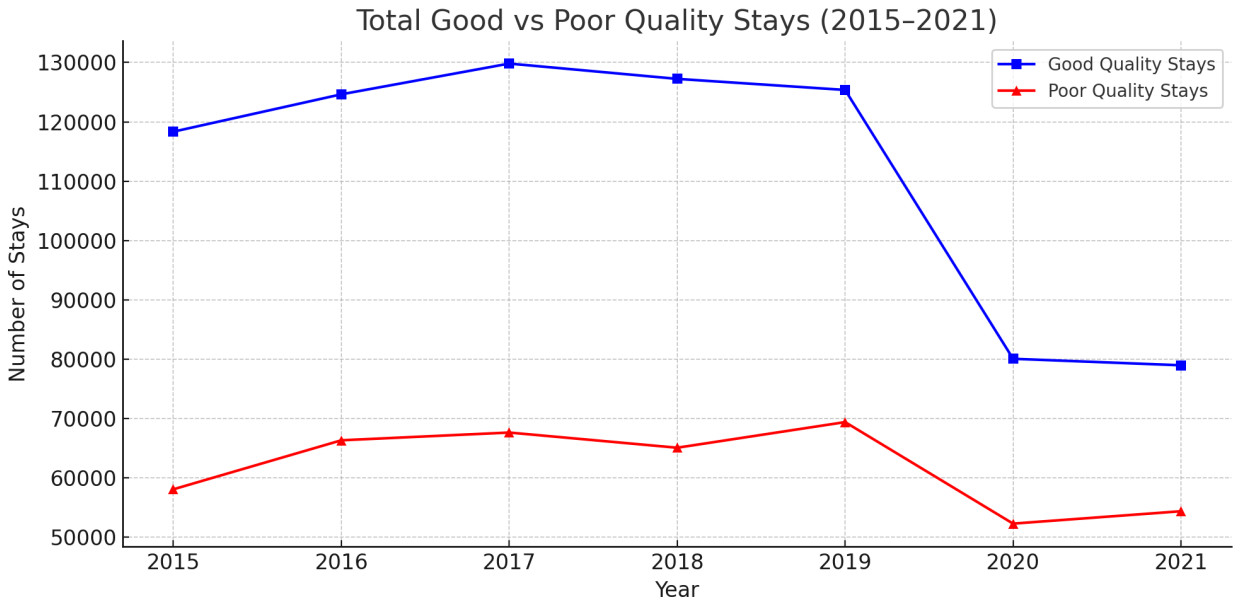


10) <u>COVID-19 Categorization Model Visuals</u>

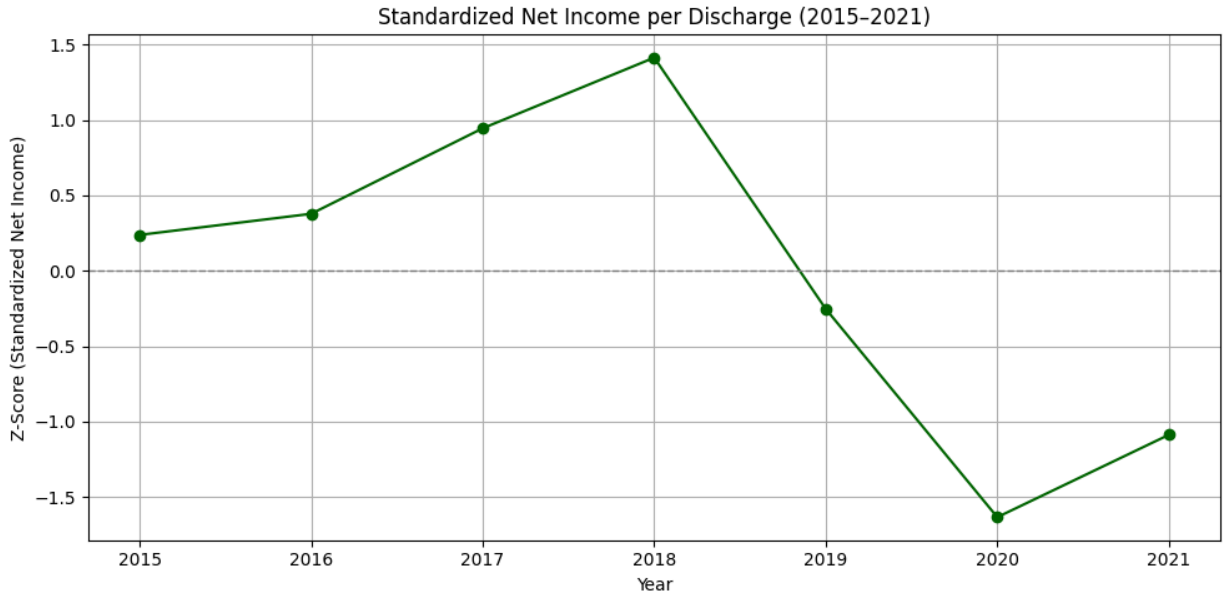




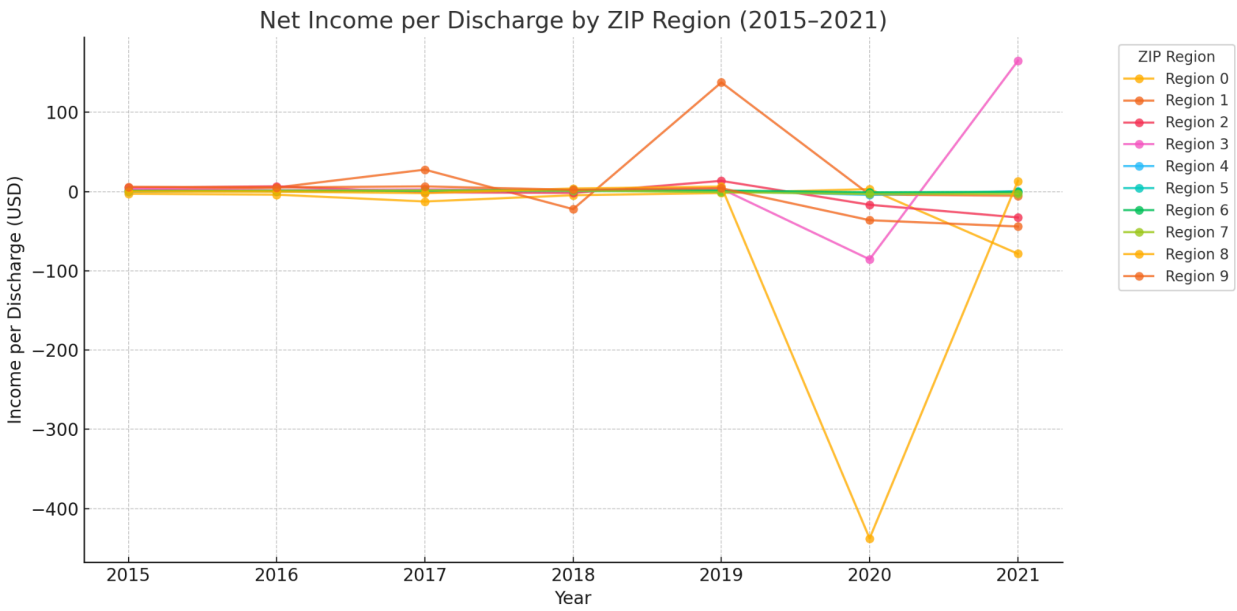
11) <u>Quality Stays</u>



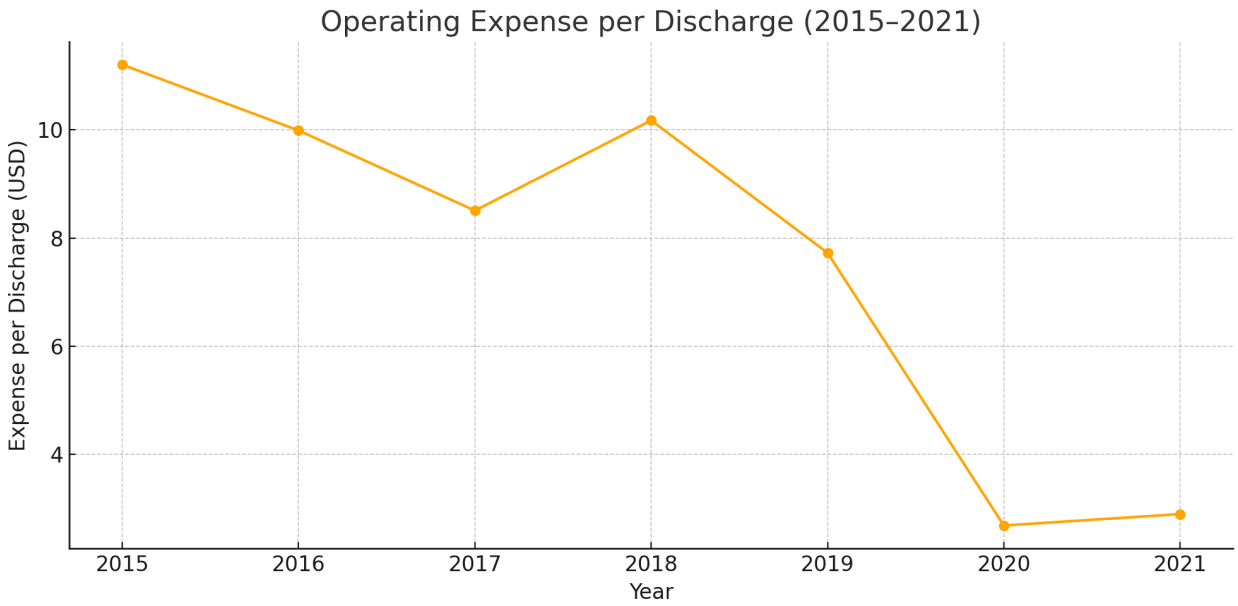
12) <u>Net Income Per Discharge by Year</u>



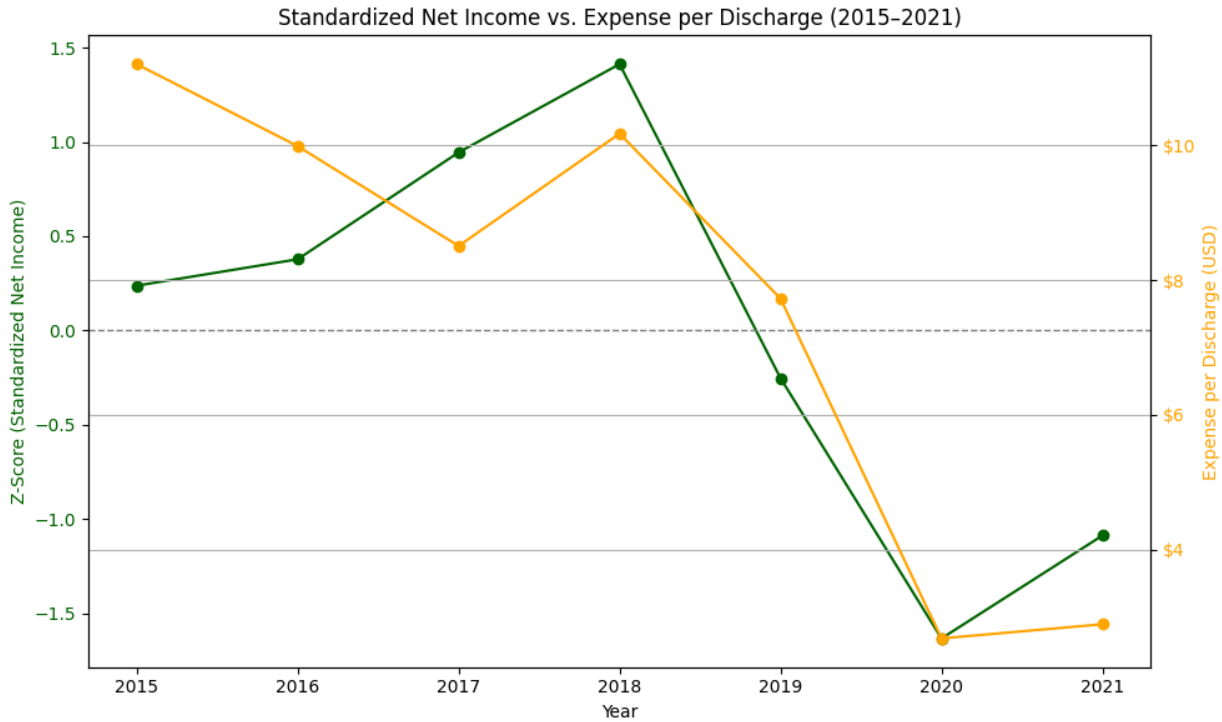
13) Net Income Per Discharge by Region



14) <u>Operating Expenses per Discharge</u>



15) <u>Net Income vs. Expense per discharge</u>



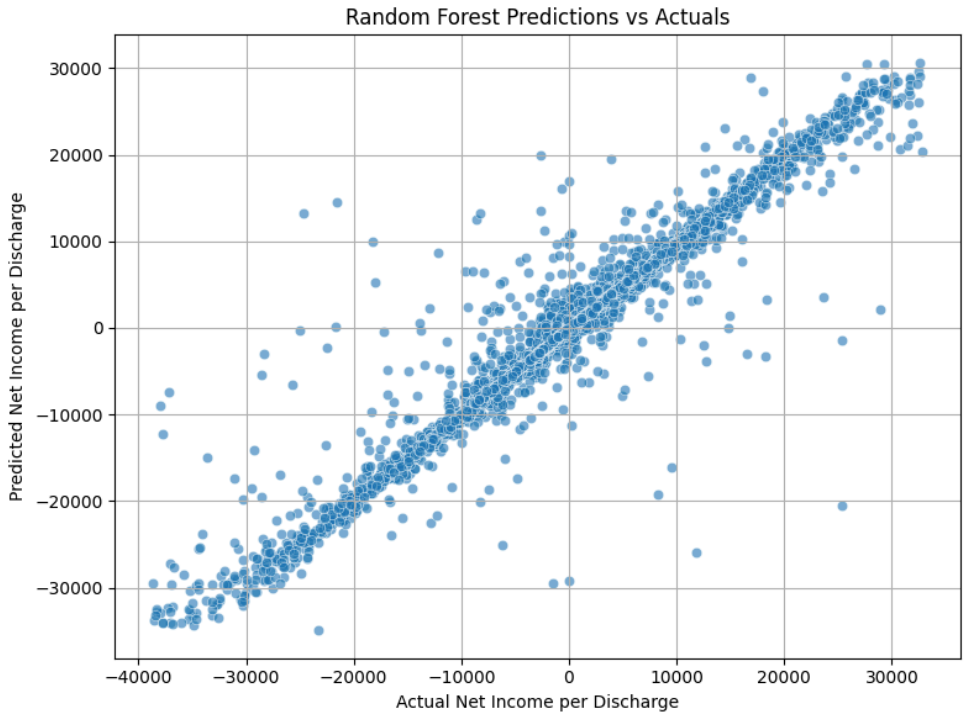
16) <u>OLS Regression Results</u>

```
Final Model Summary:
                            OLS Regression Results
==============================================================================
Dep. Variable:     Net_income_per_discharge   R-squared:                       0.545
Model:                                  OLS   Adj. R-squared:                  0.545
Method:                       Least Squares   F-statistic:                     5927.
Date:                      Fri, 02 May 2025   Prob (F-statistic):               0.00
Time:                              16:28:19   Log-Likelihood:            -1.2324e+06
No. Observations:                    103910   AIC:                         2.465e+06
Df Residuals:                        103888   BIC:                         2.465e+06
Df Model:                                21
Covariance Type:                  nonrobust
```

```
                                    coef    std err          t      P>|t|      [0.025      0.975]
-------------------------------------------------------------------------------------------------
const                             0.9482    106.220      0.009      0.993    -207.242     209.138
Total_General_Inpatient_Revenue -4419.2761    252.239    -17.520      0.000   -4913.662   -3924.890
Total_Discharges_Title_Other    -5741.0143   1834.995     -3.129      0.002   -9337.581   -2144.448
Total_Discharges_Title_XIX       -906.5340    116.360     -7.791      0.000   -1134.599    -678.469
SNF_Discharges_Title_XVIII       1527.1234    555.317      2.750      0.006     438.710    2615.537
SNF_Discharges_Title_Other       4537.0894   1825.075      2.486      0.013     959.967    8114.211
SNF_Discharges_Total            -8188.4575   1534.577     -5.336      0.000   -1.12e+04   -5180.708
Total_Discharges_Total           6606.5439   1526.484      4.328      0.000    3614.656    9598.432
Net_Patient_Revenue             -1.045e+04    414.621    -25.215      0.000   -1.13e+04   -9642.125
Inpatient_Revenue               -4137.6850    588.702     -7.028      0.000   -5291.533   -2983.838
Total_Costs                     -5796.3000    190.120    -30.488      0.000   -6168.933   -5423.667
Total_Income                     2616.9956    112.846     23.191      0.000    2395.819    2838.172
Overhead_Non_Salary_Costs        -886.7964    243.469     -3.642      0.000   -1363.993    -409.600
Gross_Revenue                    8631.0766    566.503     15.236      0.000    7520.739    9741.415
Less_Total_Operating_Expense     1.917e+04    437.775     43.792      0.000    1.83e+04       2e+04
Salaries_wages_and_fees_payable -1344.2210    138.143     -9.731      0.000   -1614.980   -1073.462
Buildings                         743.8081    156.125      4.764      0.000     437.806    1049.810
Wage_related_Costs_core           474.2736    165.185      2.871      0.004     150.512     798.035
Revenue_per_discharge            4.733e+04    367.080    128.928      0.000    4.66e+04     4.8e+04
Cost_per_discharge               4358.7780    155.166     28.091      0.000    4054.654    4662.902
Wages_per_discharge               598.0983    135.286      4.421      0.000     332.939     863.258
Operating_expense_per_discharge -8.177e+04    341.633   -239.361      0.000   -8.24e+04   -8.11e+04
==============================================================================
Omnibus:                   379383.084   Durbin-Watson:                   1.957
Prob(Omnibus):                  0.000   Jarque-Bera (JB):    1157329269929.719
Skew:                          76.277   Prob(JB):                         0.00
Kurtosis:                   16351.824   Cond. No.                         62.1
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

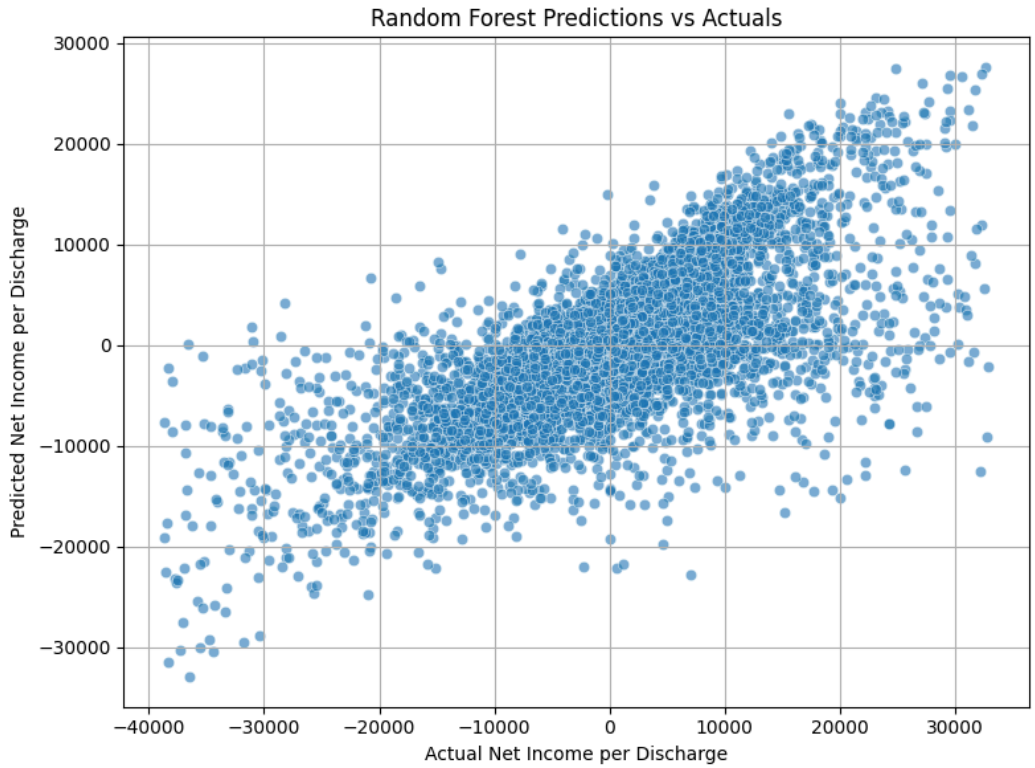
17) Random Forest Analysis: Version 1

```
Random Forest Performance:
Mean Squared Error: 2067918.53
R-squared Score: 0.9508

Top 10 Important Features:
                            Feature  Importance
13                     Total_Income    0.671797
8            Total_Discharges_Total    0.189909
22  Operating_expense_per_discharge    0.050606
7               SNF_Discharges_Total    0.047523
20            Revenue_per_discharge    0.007510
9          Net_Income_from_patients    0.003816
17  Salaries_wages_and_fees_payable    0.002900
21               Wages_per_discharge    0.002608
18                         Buildings    0.002366
10               Net_Patient_Revenue    0.002318
```



18) <u>Random Forest Analysis: Version 2</u>

```
Random Forest Performance:
Mean Squared Error: 19441811.87
R-squared Score: 0.5299

Top 10 Important Features:
                          Feature  Importance
8          Net_Income_from_patients    0.337736
18            Revenue_per_discharge    0.095263
7              SNF_Discharges_Total    0.077640
17          Wage_related_Costs_core    0.044380
9               Net_Patient_Revenue    0.043632
16                        Buildings    0.038889
20  Operating_expense_per_discharge    0.038696
11                      Total_Costs    0.035166
12        Overhead_Non_Salary_Costs    0.033421
15  Salaries_wages_and_fees_payable    0.032709
```



References

[Centers of Medicare and Medicaid Services (CMS)]